

Both hardware and software must be simultaneously considered during the design phase of any WSN. The factors to consider include transmit power, receiver sensitivity, antenna types, and physical, data link, network and transport layers

module is interfaced with the main CPU through some kind of bus, e.g., SPI. PC is used to handle the PHY and MAC layers in some cases, and the networking and transport layers in other cases. Generally, it is a two-chip solution available from different manufacturers, and hence needs two different sets of tools for firmware upgrade and software download.

Repeaters/routers. Repeaters/routers may be used to extend the range of the network, manage local address allocation and de-allocation, and participate in multi-hop routing messages.

Coordinator. It is used to coordinate the wireless communications traffic with the other elements forming the network and back to the central control point. Setting up the entire network, its maintenance and management, and allocating network addresses are the main functions of the coordinator, all of which require large amounts of computation. Hence high-performance (relative to the end device or router in the network) controllers with more powerful computing and processing capabilities must be employed.

The 32-bit MCU undoubtedly has the advantage, but 8-bit and 16-bit MCUs can also be used. Though the performance and speed of 8-bit and 16-bit MCUs is increasing continuously, the inherent bus width still limits their computing and processing capabilities. In terms of cost, the price of the 32-bit MCU has gradually dropped, and some low-end 32-bit MCUs are even cheaper than many 16-bit MCUs.

Another factor to consider is embedded memory space. Because the coordinator is the central node of the network, the protocol stack software takes up much space. Typically, it takes more than 40 kB of Flash memory and 2 kB of RAM. If less memory space is available,

the space left for user data and applications becomes too small, forcing users to reduce and optimise their codes.

Embedded processors are a better option than MCUs, as MCUs don't offer enough on-chip memory to drive all the required low-level drivers and protocol stacks, and neither do they have the potential to run an RTOS. This requires 1-4 MB of memory, which is more than available with 8-/16-bit MCUs.

When selecting a controller for the network coordinator, always consider performance over cost and power, because end devices and routers that make up the main components for a WSN are mostly low-priced.

Antenna. There are multiple antenna technologies to consider, each with its own price, performance and packaging tradeoff. Quarter-wave whip antennae offer the best benefits and radiation patterns, but are expensive. Ceramic patch antennae offer hemispherical radiation patterns, good gain and direct PCB mounts. Ceramic chip antennae are very low-cost, but low-gain. PCB trace antennae require minimal space and gain. An externally mounted antenna is a separate unit connecting to the module through an RF connector of some type, e.g., an SMA connector.

In WSN, sensor nodes will usually be battery-powered. Therefore it is imperative to optimise their functions—especially power transmission management—and make them more efficient for longer battery life. Increasing the antenna's directive gain increases the range and thereby lowers the power requirement for each transmission. This might be a bit difficult with a single antenna element. To achieve greater directivity, antenna arrays might be needed.

Receiver sensitivity. Receiver sen-

sitivity is the minimum power, in decibels (dBm), at which a radio can reliably receive data. The 802.15.4 standard specifies a minimum receiver sensitivity of -85 dBm for 2.4GHz radios and -92 dBm for 900MHz radios. All vendors of 802.15.4 radios exceed these standards, offering radios with receiver sensitivities that range between -90 dBm and -100 dBm.

Although 6 dBm may not seem like much, it can have an enormous impact on the 802.15.4 radio's line-of-sight range. Improving the receiver sensitivity from -94 dBm to -100 dBm effectively doubles the line-of-sight range of the radio. For example, if a radio with -94dBm receiver sensitivity has a 100-metre range, increasing that sensitivity by just 6 dBm extends the range up to 200 metres.

More importantly, higher sensitivity can reduce or eliminate the need for expensive, power-hungry power amplifiers, thereby reducing system complexity, cost and power consumption. For these reasons, engineers should select a radio with the highest possible receiver sensitivity.

Transmit power. Another important factor that drives the range of a radio is transmitting power. The higher the transmit power, the longer the radio's range. The 802.15.4 standard requires radios to have a minimum output power of -3 dBm, or 0.5 mW. Radios on the market today have output power ranging from 0 dBm (1 mW) to 3 dBm (2 mW).

Currently, standard and high-power modules are available from different vendors. The standard modules cover standard distances for 802.15.4 up to 100 metres. High-powered modules offer much more transmission power and cover five or six times the distance of the normal power modules. Depending on the terrain and visibility, the distances of high-power devices could reach up to 500 metres or more.

To be concluded next month

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